

Updated Design

After adding all the important improvements and additional components, the final notification system becomes:

- More scalable
- More reliable
- More secure
- Easier to maintain
- Better monitored

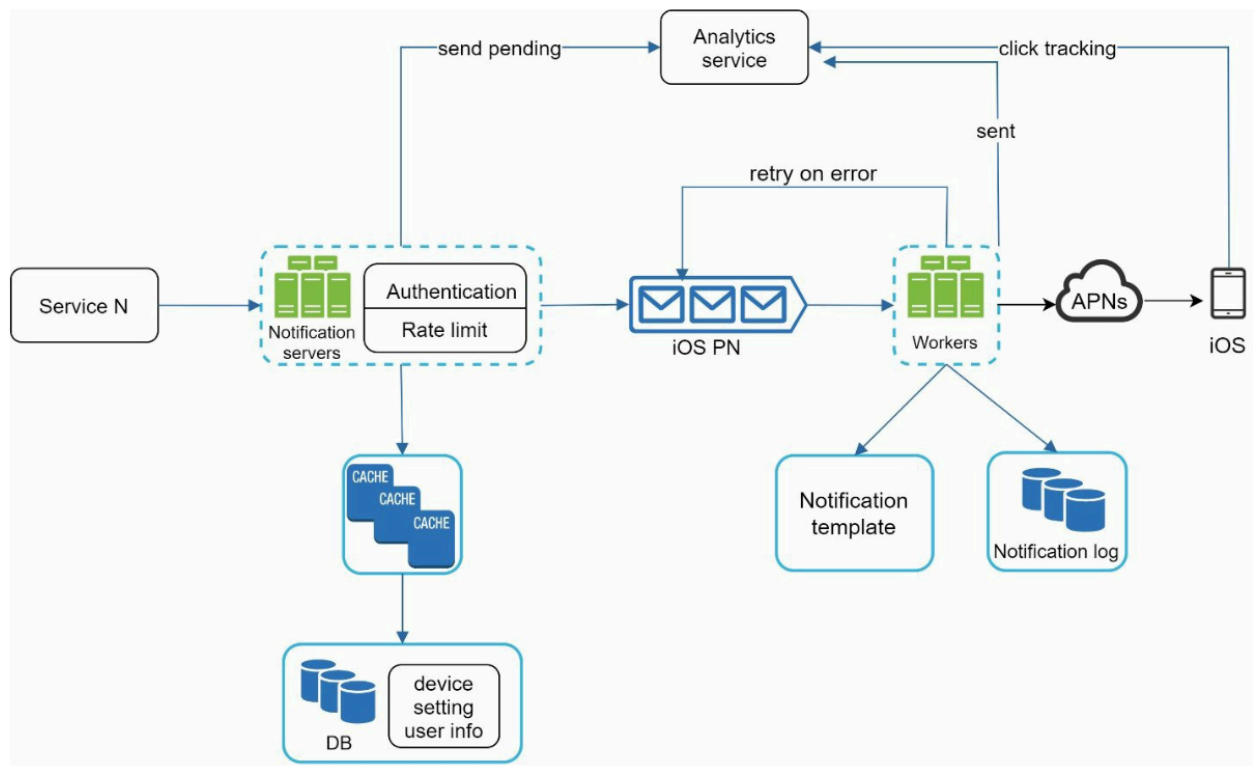


Figure shows the complete updated notification system architecture.

Improvements in the Updated Design

Compared to the previous design, several new components and capabilities are added.

The major additions include:

- Authentication
- Rate limiting
- Retry mechanism
- Notification templates
- Monitoring system
- Tracking system

1. Authentication

The notification servers now include:

Authentication Feature

Purpose

Authentication ensures:

None

Only authorized and verified clients can send notifications.

Why Authentication is Important?

Without authentication:

- Attackers may send fake notifications
- Spam notifications may be generated
- APIs may be abused

Benefits

- Improves system security
- Prevents unauthorized access
- Protects notification APIs

2. Rate Limiting

Notification servers are also equipped with:

Rate Limiting Feature

Purpose

Prevent sending excessive notifications within a short period.

Why Rate Limiting is Needed?

Too many notifications can:

- Overwhelm users
- Reduce engagement
- Cause users to disable notifications

Example

Limit:

- Push notifications per hour
- Emails per day
- SMS alerts per minute

Benefits

- Prevents spam behavior
- Protects users from notification overload
- Improves user experience

3. Retry Mechanism

The updated design introduces:

Retry Mechanism

to handle notification delivery failures.

Failure Scenario

Failures can happen because:

- APNS is unavailable
- FCM timeout occurs
- SMS provider fails
- Network issues happen

Retry Flow

1. Notification sending fails
2. Notification is placed back into the message queue
3. Workers retry delivery later

Retry Attempts

Workers retry notifications:

None

For a predefined number of times

Benefits

- Improves reliability
- Prevents notification loss
- Handles temporary third-party failures

4. Notification Templates

The updated design also includes:

Notification Templates

Purpose

Provide a reusable and standardized notification creation process.

Benefits

1. Consistent Notification Format

All notifications follow the same structure.

2. Faster Notification Creation

Developers reuse templates instead of creating messages repeatedly.

3. Reduced Errors

Templates reduce formatting mistakes.

4. Easier Maintenance

Updating templates updates multiple notifications automatically.

5. Monitoring System

A monitoring system is added for:

System Health Checks

Purpose

Monitor:

- Queue size
- Worker performance
- Failure rate
- Processing delays
- System load

Benefits

- Detect bottlenecks
- Identify failures early
- Improve operational visibility

Example

If queue size increases rapidly:

- More workers can be added

to prevent notification delays.

6. Tracking System

The updated design also integrates:

Event Tracking and Analytics

Purpose

Track notification-related events such as:

- Sent notifications
- Delivered notifications
- Open rates
- Click rates
- User engagement

Benefits

- Understand customer behavior
- Improve notification effectiveness
- Optimize campaigns

Updated Design Summary

Added Component	Purpose
Authentication	Secure notification APIs
Rate Limiting	Prevent excessive notifications
Retry Mechanism	Handle delivery failures
Notification Templates	Reusable notification formatting
Monitoring System	System health monitoring

Tracking System	Analytics and event tracking
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Final Updated Architecture

The updated architecture now supports:

- Scalability
- Reliability
- Security
- Fault tolerance
- Monitoring
- Analytics
- Better user experience

Key Takeaways

- Authentication secures APIs
- Rate limiting prevents notification flooding
- Retry mechanism improves reliability
- Templates standardize notifications
- Monitoring improves system health visibility
- Tracking enables analytics and optimization